

The empirical recurrence for OEIS A236776: recovering a conjecture that is not written down

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Abstract

OEIS A236776 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 474$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236776 is “Number of $(n+1) \times (2+1)$ 0..2 arrays colored with the maximum plus the upper median plus the minimum minus the lower median of every 2×2 subblock.”. It has offset 1 and begins

474, 4170, 36260, 315666, 2750070, 24023328, 209304200, 1823484380, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:31:55, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 474$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 474$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 948$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	0	c_{17}	−1209810443	c_{33}	1129112805388	c_{49}	−402007494852
c_2	6	c_{18}	1678490182	c_{34}	−464132316544	c_{50}	−974054829802
c_3	93	c_{19}	3687106287	c_{35}	529749214776	c_{51}	1427594945616
c_4	2955	c_{20}	−610089632	c_{36}	2072036217639	c_{52}	−139308875464
c_5	9582	c_{21}	5871193817	c_{37}	−183692580139	c_{53}	423320906368
c_6	31395	c_{22}	−20950469795	c_{38}	−1859797761966	c_{54}	221394819488
c_7	86930	c_{23}	−29892846560	c_{39}	−1000325521749	c_{55}	−425036663488
c_8	−352833	c_{24}	68684584399	c_{40}	−801551318263	c_{56}	−307871494144
c_9	−1318187	c_{25}	−47809512364	c_{41}	−2653174807559	c_{57}	−51864194816
c_{10}	−1950367	c_{26}	−31808504368	c_{42}	2176001972308	c_{58}	50583256064
c_{11}	−4491068	c_{27}	327473430725	c_{43}	3874519660229	c_{59}	−10695155712
c_{12}	26135112	c_{28}	−229061395503	c_{44}	−1466981746903	c_{60}	−15472605184
c_{13}	78757764	c_{29}	−218962845119	c_{45}	2483063439241	c_{61}	256327680
c_{14}	−53299604	c_{30}	527507053921	c_{46}	372704231412	c_{62}	−321257472
c_{15}	−104713741	c_{31}	−884561663328	c_{47}	−4302496797622		
c_{16}	−293115248	c_{32}	−467070391367	c_{48}	1196790555810		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

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